

Week 1 - What will we build and how will we judge it?

Artificial Intelligence in FinTech | Academic year 115, semester 1

M0L23910 | M0L20A, second-year master's class, Information Management

3 credits | 3 hours per week | Instructor: Wen-Yi Lee

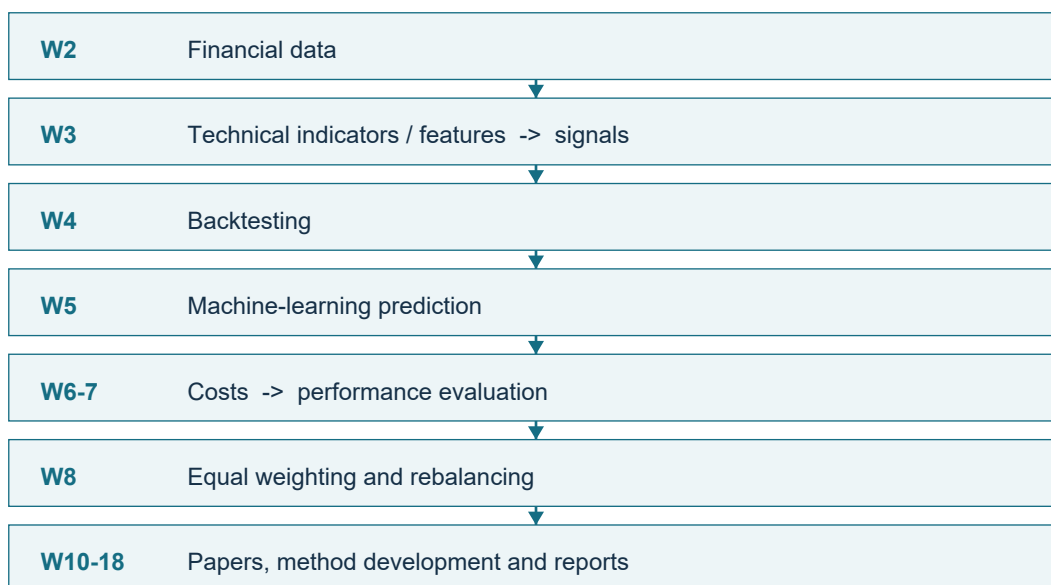
The course in one question

Can we turn an investment idea into a trading or portfolio decision, then check whether the evidence supports it?

We begin with data and simple trading rules. Later, machine learning predicts an outcome, and a weighting method decides how much money goes to each asset. A prediction is not yet a trade, and a trade is not automatically profitable.

What you will learn

- Prepare financial data and explain what a calculated return means.
- Build and check a strategy using information available before its trades.
- Compare weighting methods, develop a modification, and explain its evidence and limitations in English.



The diagram shows why we do not begin by searching for a profitable AI model: its inputs, trades and evaluation must be defined first. When a weighting rule changes, its costs and results must be checked again.

Today, follow one AAPL example through three questions: what are we comparing, which price changes belong to the account, and what does the result support? No coding or model fitting is required.

Issue 1 - What exactly are we comparing?

"This strategy is better" is incomplete. Better could mean more money at the end, smaller losses along the way, or fewer trades. Choose one question before looking at the result.

One question for this example

Does a fixed AAPL trading rule finish with more money than buying and holding AAPL over the same period?

The trading rule uses closing prices. After a Close above the Close five price intervals earlier, it requests full investment in AAPL; otherwise it requests cash. The change is assumed to take place at the next stored Close. We will implement this rule in Week 4, not today.

Account	Decision
Trading rule	Switch between AAPL and cash.
Buy and hold	Buy AAPL at the start and keep it.

Both start with USD 1,000 on January 2, 2024 and are valued through June 28. The trading rule stays in cash until enough history is available and a signal can be executed. Buy and hold is chosen in advance and buys at the first Close.

For this first comparison, fractional shares are allowed. Costs, dividends, taxes and cash interest are excluded. No new trade is added at the final Close. These are hypothetical account rules, not observed orders. The reported values therefore describe gross, price-only results, not take-home investment profit.

Before seeing the result

If the trading-rule account finishes below buy and hold, would its positive dollar gain alone answer our question in its favor?

Discuss

1. A classmate changes the question to "Does the rule trade less often?" What must be counted instead of comparing only the final account values?
2. One team reports results before costs and another after costs. Before comparing them, which calculation conditions must be made consistent?

Issue 2 - Did our account earn that price change?

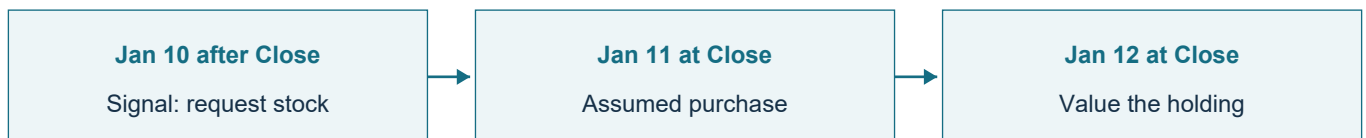
A stock price can rise while an account is still in cash. To explain an account's gain, first identify when it starts holding the stock under the assumed rule.

Read four actual AAPL closes

These are Yahoo Finance Close observations accessed through yfinance, rounded to cents. They are prices in USD per share, not evidence of our order executions.

Date	Close (USD/share)
2024-01-03	184.25
2024-01-10	186.19
2024-01-11	185.59
2024-01-12	185.92

January 10 is five stored price intervals after January 3. Its higher Close gives a stock signal under the rule. But that signal is calculated after the Close: the rule assumes a purchase at January 11 Close, not January 10 Close.



For one assumed share, the January 11-to-12 holding gains about USD 0.33 before costs: the price moves from USD 185.59 to USD 185.92. The earlier move from USD 184.25 to USD 186.19 helped form the signal; it is not a gain earned by that new holding.

A published Close does not guarantee an actual order would fill there. Here the purchase date and price are assumptions. Week 4 will make the holding dates explicit, and Week 6 will examine execution and costs.

Discuss

1. Why may this January 10 signal not claim the gain from January 3 to January 10? Identify the assumed purchase date and the two prices that belong to its first holding interval.
2. If the purchase happens only after January 12 Close, may the account still claim the USD 0.33 gain? Explain which holding interval would need to change.

Sources: [Yahoo Finance AAPL history](#); [yfinance price history](#); [Investor.gov order types](#).

Issue 3 - What does the result actually support?

Return to the question: does the fixed AAPL trading rule finish with more money than buy and hold? Compare the two accounts, not just the trading rule with its own starting balance.

Read the result before choosing the claim

The table previews the executed Week 4 calculation using the same AAPL observations and the rules stated in Issue 1. Prices are real observations; account values are computed under assumed execution. Both accounts exclude costs, dividends, taxes and cash interest.

Valuation date	Trading rule (USD)	Buy and hold (USD)
2024-01-02	1,000.00	1,000.00
2024-06-28	1,068.80	1,134.56

The rule gains USD 68.80, but buy and hold gains USD 134.56. The rule finishes USD 65.76 lower. A positive gain therefore does not support the claim that the rule outperformed this benchmark on final value during this sample.

This table does not tell us whether either account fell below USD 1,000 before June 28. Nor does it establish what either account will earn in a later period. Those are different questions requiring additional evidence.

Your turn: ask a different, answerable question

Keep AAPL, the dates and the two rules fixed. Ask whether either account was worth less than its initial USD 1,000 on an earlier date. Describe which dated account values you would inspect and what observation would answer yes. No risk formula is needed yet.

Discuss

1. Which two values contradict the claim "The rule ended with more money than buy and hold"? Rewrite the claim so it states what this sample actually shows.
2. Can the two final balances establish that neither account ever fell below USD 1,000? Explain which earlier observations are needed before making that claim.

The activity is a classroom discussion, not an additional report or form to submit. The purpose is to connect a question to the evidence that could answer it.

Calculation reference: [Week 4 - From Signals to a Backtest](#). Week 2 first explains the price and return fields; Week 7 later develops risk and performance evaluation.

How the course is assessed

Grading

- Class participation and weekly learning evidence - 40 points
- Report 1 - Quantitative Trading Report: 15 points
- Report 2 - Weighting Method Progress Report: 15 points
- Report 3 - Final Report: 30 points

Three reports, one developing project

Session	Main purpose
Week 10: Report 1	Connect prediction, trading, backtesting and an initial weighting approach.
Week 14: Report 2	Explain a weighting-method modification and preliminary evidence.
Weeks 16-17: Final	Present and defend the completed comparison.

Students may develop their own way to turn predictions or signals into portfolio weights. The proposed change must be clear enough to compare with an existing method. Detailed report requirements will be supplied for the report sessions; today's AAPL example is not a prescribed project solution.

Submit through GitHub

Reports are submitted through GitHub. The main report, presentation and oral explanation use English. At each deadline, the team identifies one Git commit as the frozen version for presentation and evaluation. Later commits do not replace it. Keep the original version and explain how any later correction changes results or conclusions.

Present, compare and explain

Students compare and rank other teams, not their own, using stated criteria and common financial evidence. Classmates' rankings do not directly determine team grades. Individual reviews, rankings and grading-related personal records remain private.

Do not publish credentials, restricted data, personal information or grading records. Approved accommodations follow the university's official procedure while preserving the assessed learning outcomes.

Semester at a glance

The first part builds the trading workflow. The later part uses papers and reports to develop and evaluate weighting methods. The instructor will announce the papers in class or through the course GitHub repository.

Week	Focus
01	Course direction and testable investment questions
02	Financial data and returns
03	Technical indicators, features and signals
04	Positions, trades and backtesting
05	Machine-learning prediction and chronological evaluation
06	Transaction costs, execution delay and turnover
07	Performance, risk and backtest validity
08	Equal weighting, rebalancing and diversification limits
09	No class: instructor travel
10	Report 1
11	Papers related to weighting methods
12	Papers related to weighting methods
13	Papers related to weighting methods
14	Report 2
15	Papers related to weighting methods
16	Final reports: first session
17	Final reports: second session and peer comparison
18	Papers related to weighting methods

No single textbook is required. Instructor-prepared materials, open or authorized financial data, and assigned papers support the course. Books and papers support financial methods; company, exchange and provider records support market facts; software documentation supports code behavior. None alone establishes that a classroom strategy will make money.

Next: [Week 2 - Financial Data and Returns](#). Start by identifying what each price, date and unit means.

Course reference - Educational use and GenAI

These rules apply throughout the semester. They are collected here so that administrative requirements do not interrupt the three teaching questions.

Educational use

Course examples, strategies, backtests and discussions are for education, not investment advice, solicitation, a research report or a trading instruction. Do not use a classroom example to trade real money. Actual investment decisions remain the investor's responsibility and may require qualified professional advice.

Backtests do not represent future performance. Leakage, unavailable future inputs, survivorship bias, underestimated costs, slippage, liquidity limits, overfitting, parameter selection, changing markets, short samples and unrealistic fills can affect them. A classroom return or risk result is conditional evidence, not a profit guarantee.

Permitted uses of GenAI

GenAI may help draft a hypothesis, explain an error message, organize feature descriptions, revise overconfident wording, or suggest questions and risks to verify. It does not replace judgment or verification.

Do not submit unreviewed output, conceal material GenAI use, fabricate data, results, execution records or literature, present generated investment advice as reliable, or use unchecked generated code or financial explanations. Do not enter credentials, personal or restricted data, or another student's work into a GenAI service.

Responsibility and disclosure

Students remain responsible for their calculations, sources, code, interpretation and oral answers. If GenAI affects a submitted report, code, figure, table or presentation, identify the tool, purpose, main prompt, retained content, removed or changed content, and human verification performed.

During oral questions, explain the work yourself; do not ask GenAI to answer in real time. Disclosure does not replace verification.

Course reference - Claims and report disclosures

Say what the evidence supports

Do not claim guaranteed profit, proof of a market pattern, the best strategy, or continuation of a result into the future without the necessary evidence.

Instead, state the observed result, its conditions and its limits. For the AAPL example: the trading rule ended above its starting balance but below buy and hold over January 2-June 28, 2024, with costs and dividends excluded. This is a comparison within one sample, not a claim of future performance.

Required information in strategy reports

Each strategy report must identify:

- The data source and period.
- The target and holding period.
- The benchmark.
- Transaction-cost assumptions.
- Maximum drawdown or another relevant downside-risk measure.
- Whether information leakage was checked.
- Whether cost or threshold sensitivity was examined.
- The parts affected by GenAI.
- Why the classroom evidence is insufficient for direct real-money trading.

These are existing course requirements, not extra submissions for the three Week 1 questions. The technical terms will be developed in the relevant weeks.

The goal is to define a hypothesis, build data and a model, check bias, backtest and stress test, disclose limitations, and explain a decision using evidence. Finding one high historical return is not the goal by itself.

Public course materials: [FinTech course directory](#). Papers without distribution rights, instructor answers, personal EMI documents and grading records are not published there.